

Lab Assignment 1

EE5518 – AY2025/2026 Sem1

Submission Deadline: 13th October'25 11:59AM

Lab experience 1: Getting started with Cadence Virtuoso via Interconnect Modelling and Simulation.

Consider a 2-core processor with 2 dedicated GPUs and 2 DSP cores in 45nm operating at 1V and max frequency of 200MHz. The path to each block from the *RESET PIN* is shown in Fig. 1 along with the corresponding global/intermediate (solid-black/dotted-blue) interconnect lengths. The interconnects are assumed to be drawn with minimum widths allowed by the technology node. Using an ideal voltage source V_{RST} and a $3k\Omega$ resistor at the *RESET PIN*, reset signal is triggered at $t = 0$, synchronous to rising edge of clock. The input capacitance of the reset buffer in each block is shown via the dashed arrows in Fig. 1.

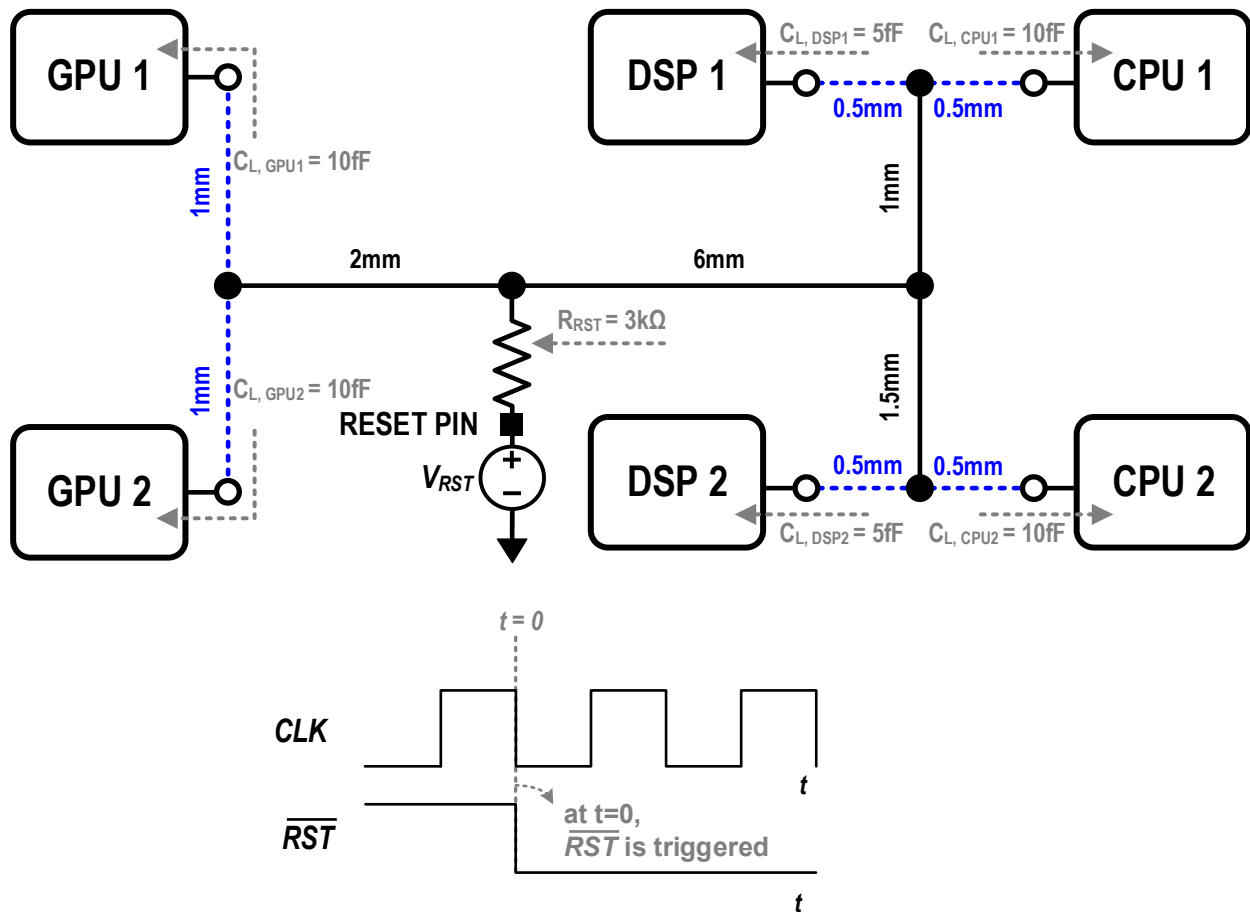


Fig. 1: Reset interconnect path in 2-Core Processor with timing diagram

Table A: Interconnect technology data in 45-nm – intermediate and global metal

	technology parameter	definition	value (intermediate wire)	value (global wire)
physical constants	r_w	wire resistance per unit length	1.20 $\Omega/\mu\text{m}$	0.05 $\Omega/\mu\text{m}$
	c_w	wire capacitance per unit length	0.13 fF/ μm	0.19 fF/ μm

Assume that the rise and fall delay for all signals is 0 and there is no coupling capacitance between the interconnects. It is also assumed that clock is ideal and there is no clock skew between the blocks. You may refer to the technology parameters in Table A.

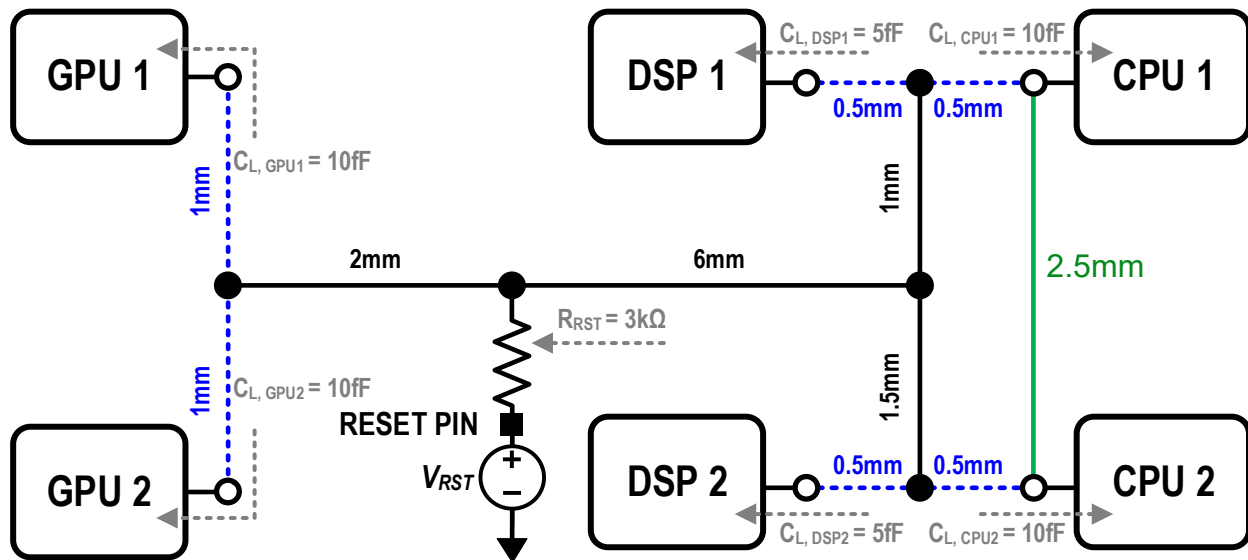


Fig. 2: Modified reset interconnect path in 2-Core Processor for Q4. & Q5.

- Q1. Draw the equivalent circuit for the described interconnect path in Fig. 1. Annotate all the R, C values on the equivalent circuit, and estimate the **propagation delay** from *RESET PIN* to each receiver module using Elmore delay method with π lumped RC model.
- Q2. Build the schematics and testbench for the equivalent circuit in Q1. Simulate your testbench and verify the hand calculation results in Q1.
- Q3. Assuming that Recovery and Removal time are 0, check if reset signal can be treated as a synchronous signal and no further optimization is needed.
Hint: check if all blocks receive the reset signal before the next rising edge of the clock for reset signal, and hence the latter can be treated as synchronous (or not).
- Q4. Assuming that there is one extra global interconnect (as shown in Fig. 2 in green) which form a mesh interconnect path. Estimate the **propagation delay** from *RESET PIN* to each receiver module using Elmore delay method with π lumped RC model.
(Hint: you may refer to the lecture notes for Elmore delay estimation with loops)
- Q5. Verify your estimations in Q4. through simulation, and comment on the simulation results.

(*Please show all the calculation steps & relevant screenshots/figures. No marks will be given for result/value-only answers without any explanation/step.)

Experience 2: Designing simple standard cells & digital circuit, and cleaning up DRC & LVS.

In this lab experience, you are required to design simple standard cells (INV, NAND2/NOR2) using standard cell method (refer to lecture notes) in 45nm while adhering to the design requirements showing in Table B, then assemble your own standard cells into a ring oscillator (RO), which is a simple & widely used digital circuit for periodic signal (e.g., clock) generation, or as process monitors to judge if a particular chip is faster or slower than nominal expectation.

(*x* in Table B/C is the second last character of the matriculation # of the first student in group as per Canvas)

Table B: design requirements for your standard cells

cells	INV (all) NAND2 (for $x \leq 4$) NOR2 (for $x \geq 5$)
*standard cell height	10T (for $x \leq 4$) 12T (for $x \geq 5$)
standard cell width	minimize the cell width while meeting design rules and other constraints
top metal layer	metal 1 (M1)
pn ratio	1.4
channel length	minimum length
finger width	maximize the finger width while meeting design rules and other constraints
number of fingers	10 (for $x \leq 4$) 6 (for $x \geq 5$)
pin position	M1
VDD/VSS rails	yes (each individual cell should have the rails)

*represent in terms of number of metal tracks (1T = 1 M1 pitch = 1 min. M1 width + 1 min. M1 space)

Table C: number of inverters in your ring oscillator

x	y (# of INV cells)
0, 2, 7	14
1, 3, 6, 9	16
4, 5, 8	18

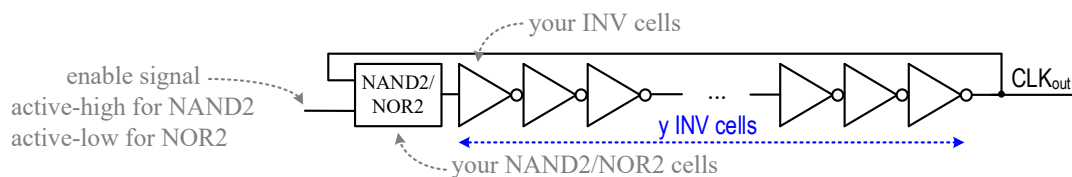


Fig. 3 schematics of a simple ring oscillator

- Q1. Show schematics & layout^(a) of your INV cell, and the corresponding DRC/LVS results^(b).
- Q2. Show schematics & layout^(a) of your NAND2/NOR2 cell, and the corresponding DRC/LVS results^(b).
- Q3. Compose the schematics of your RO based on Table C & Fig. 3. Simulate the behavior of your RO by feeding a rising/falling edge to the NAND2/NOR2 gate.
- Q4. Show the layout & the DRC/LVS results^(b) for your RO. (You may use metal 1-3 to connect the pins.)
- Q5. Report the frequency, rising time, and falling time of the generated clock (CLK_{out} in Fig.4) from your RO. Analyze the factors that may affect the three values.
(Hint: this is a pre-layout simulation without layout parasitic, you will explore more in Experience 4.)

^(a)Screenshots that clearly show the **transistor sizing** in schematics, and measurements of **cell height & width** in layout via rulers. ^(b)You will need to add taps for LVS.

Experience 3: Characterizing technology parameters in 45nm using your own cells

In this lab experience, you are going to characterize the p_n ratio, FO4, and intrinsic delay (τ) for your own cells using the following circuit template (replace the inverters with the required cells in each question):

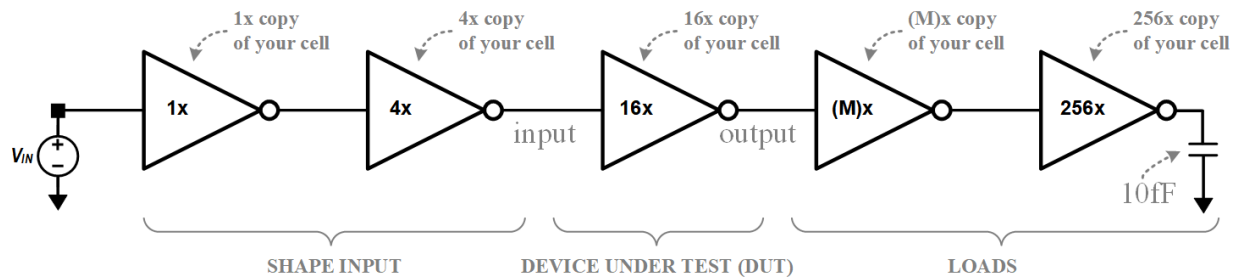


Fig. 4: Circuit template to be used

Table D: cells for Q5.

x	cell
0, 2, 4, 6, 8	NAND3
1, 3, 5, 7, 9	NOR3

Table E: sizing method for Q5.

x	method
0, 2, 4, 6, 8	high speed method
1, 3, 5, 7, 9	standard cell method

- Q1. Set $M=64$. Plot p_n ratio (W_p/W_n , from 1 to 2 in 0.2 step) vs rising/falling delay ratio at 1V. You may set W_n to a fixed value (e.g., from Experience.2.Q1.) and change W_p at each step.
(Hint: You may use tool's built-in calculator to measure accurate delay).
- Q2. Set $M=64$. Plot p_n ratio (W_p/W_n , from 1 to 2 in 0.2 step) vs rising/falling delay ratio at 0.5V. You may set W_n to a fixed value (e.g., from Experience.2.Q1.) and change W_p at each step.
(Hint: You may use tool's built-in calculator to measure accurate delay).
- Q3. Determine the proper value for M . Plot the simulated FO4 delay vs supply VDD (from 0.5V to 1.1V in 0.05V step) using your INV cell in Experience.2. Assume the maximum logic depth of your digital integrated circuit is 50·FO4, what is the theoretical maximum operating frequency (ignore setup/hold, uncertainty, and other variations) at 0.9V?
- Q4. Plot the measured propagation delay (τ_{PD}) vs M (from 16 to 160 in 16 step), find the intrinsic delay τ using your INV cell in Experience.2. What is the meaning of “M” in this question?
- Q5. Build a NAND3/NOR3 gate (refer to Table D.) Fill up below table using your INV cell as the template inverter, sizing method in Table E, and technology parameters in Q4. Comment on the results and clearly show all the steps & calculations.

Gate	Input	theoretical logical effort	rising logical effort (g_u)	falling logical effort (g_u)	average logical effort (g)	rising parasitic delay (p_u)	falling parasitic delay (p_d)	average parasitic delay (p)
	A							
	B							
	C							

Experience 4: Post-layout simulation

In this lab experience, you will explore the layout-induced performance downgrade.

(Hint: you may refer to “Cadence IC Design Manual revised” for post-layout simulation)

Q1. Redo Experience 2. Q5. with layout extractions. Comment on your observations.

Q2. Redo Experience 3. Q3. with layout extractions. Comment on your observations.

Survey: individual contribution

Clearly state the contribution (%) from each group member with below template:

Matriculation number (e.g., Axxxxxxx)	Contribution (%)

Report:

- Submit your assignment in the form of a report.
- Do not report the result directly. Try to explain your results and the steps followed.
- Clearly state any additional assumptions that you have taken.
- The final report should be in the pdf format only.
- **Clearly indicate the contribution (in %) of each student on the front page of the report. Intra-team marks may vary based on the contribution reported.**

Submission Instructions:

- **Name your file as:** Matric_Number_Assignment_1.pdf and Matric_Number_Assignment_1.zip including your own library of the design in each experience.
 - Ex. A1234567A_Assignment_1.pdf & A1234567A_Assignment_1.zip
- Upload to **Canvas EE5518 -> Files -> Assignment 1** before 13th October'25 11:59AM. **(submit the report and zip file separately under the same assignment tab)**
- A penalty of 25% applies for late submissions of up to 1 week. After that, the submissions will not be considered.
- **Plagiarism is penalized with a 100% penalty for all SOURCES and RECIPIENTS and will be reported for further action.**